Database Project

Choose your adventure

2025-2026

laSalle
UNIVERSITAT RAMON LLULL

INTRODUCTION

The objective of the project is to implement a database in teams of three people. The starting point is a dataset in CSV (Comma-Separated Values) format.

Stage 1: Database design

The first step is to create the **conceptual model** (entity-relationship model) of the database that allows for storing all the information from the selected dataset. This conceptual model must also be expanded with a CSV file created by the team. This new file must be linked in some way to the rest of the CSV files. To generate this file, you can use the Mockaroo tool. (https://www.mockaroo.com/) or something similar.

Based on the developed entity-relationship model, a list of at least **30 questions** that the database must be able to answer will be developed. The objective of the questions is to demonstrate that the database is well designed and that interesting information can be extracted. Examples of questions include: *Which doctors have more appointments in the afternoon? Which doctors have treated more patients than the average number of patients treated in 2025?*

The entity-relationship model must be **normalized** (up to third normal form) and follow the syntax shown in the Databases classes. The model must be accompanied by an explanation of all the design decisions made.

The model must be created using the MIRO application (https://miro.com) following the syntax shown in class. The submitted version cannot be handcrafted.

Once the entity-relationship model is complete, the **relational model will be developed** . Each team member will translate a portion of the entity-relationship model. The entire team will work together to agree on a final relational model that meets all database requirements. The final relational model must comply with third normal form. All team members must be familiar with and understand the final relational model in detail.

Expected results of stage 1:

- Entity-relationship model
- Relational model
- Database description with design decisions
- List of 30 questions to validate the database
- New CSV file created by the team

Stage 2: Implementation of the physical model

The next step will be the implementation of the relational model in a physical model for the Oracle database manager.

First, the data types for the columns (attributes) will be chosen based on the relational model. To do this, the most appropriate data type must be selected for each column based on the requirements from Stage 1. Each team member will propose the types (VARCHAR2, DATE, NUMBER, etc.) for the attributes of each table in the module they oversee. The entire team will work together to agree on a final list of types for all table attributes. The justification for the data type selection will be described in the report to be submitted.

Based on the relational model and the data type selection, the team will program an SQL script to create the model tables **(physical model)**. The result of this step is a single text file with a .sql extension that will contain all the creation commands for all the tables in the database. The script must be programmed so that it can be run multiple times without errors, and the physical model is automatically created each time the script is run.

Once the tables have been created in the Oracle database, the data must be imported. Each CSV file will be imported into the database to a temporary table. Using INSERT-SELECT scripts, the data will be moved from the temporary tables to the physical model tables. The names and contents of the CSV files cannot be modified in any way, and all the data in the CSV files should be imported to the temporary tables exactly as is. The result of this step is a single text file with a .sql extension that will contain all the data import and population scripts for all the database tables.

Expected results of stage 2:

- Database creation script
- CSV File Import Script, including the INSERT-SELECTs from the temporary tables to the physical model tables.

Stage 3: Queries and Triggers

Once the data has been successfully imported, the validation questions identified in step 1 will be implemented in SQL. **All the SQL concepts seen in class must be used for the resolution of the proposed queries (units 4 and 5).** In case the questions proposed in phase 1 do not require the use of some of the concepts seen in class, additional queries will be proposed to meet this requirement. Additionally, **at least three triggers** will be implemented to automate some part of the database, such as adding, modifying, or deleting records.

Both queries and triggers can be resolved in many ways. Therefore, your goal when solving them is to meet the requirements of the query statement by trying to reduce the complexity of query resolution and the additional tools used, such as unnecessary subqueries, malformed JOINs, or extra functions.

For each query and trigger, you must explain how the query works and demonstrate that it is properly implemented, using supporting queries to validate the results.

Expected results of stage 3:

- Script with the solution to the queries and triggers
- Detailed report of stages 1, 2 and 3

Delivery instructions

The project (all three stages) must be developed by a three-member team. All team members must be from the same class group. There will be two submissions for the project:

Phase 1

The first stage of the project (conceptual and relational models) must be submitted by **23:59 on March 22, 2026**.

The filename of the submitted file will be: **Group<#>.zip** , where **<#>** is the group number assigned by the subject instructor. This file must be a ZIP file containing the following folders:

- **Report** . This folder should include the report written by the team following the given template (available on eStudy) until stage 1 of the project.
- **Models** . This folder should include high-resolution images (png, jpg, pdf, etc.) of the entity-relationship and relational models so that they can be enlarged.

The first stage of the project can be submitted in the **second call by 23:59 on May 17, 2026**. **The maximum grade for this call is an 8** .

Phase 2

The final submission of the project must include all the stages of the project with the corresponding corrections done on the first stage of the project based on the feedback received by the subject instructors. All the project documentation must be submitted by **23:59 on May 17, 2026**.

The filename of the submitted file will be: **Group<#>.zip** , where **<#>** is the group number assigned by the subject instructor. This file must be a ZIP file containing the following three folders:

- **Report** . This folder should include the report written by the team following the given template (available on eStudy).
- **Models** . This folder should include high-resolution images (png, jpg, pdf, etc.) of the entity-relationship and relational models so they can be enlarged.

- **Scripts**. Three scripts should be included in this folder.
  - A script to create the physical model (For example: *G1_physicalmodel.sql*, where <1> corresponds to your group number)
  - A script to import the CSV data (For example: *G1_import.sql*, where <1> corresponds to your group number)
  - A script with the queries and triggers (for example: *G1_queries.sql*, where <1> corresponds to your group number). This script must follow the template provided in eStudy.
- **Presentation.** A presentation in **pptx/pdf** to explain the project in public.

The maximum grade for any submission of a phase done in the extraordinary call is a 6.

Important considerations:

- **If a file is missing, the submission will not be accepted.**
- **If the report template is not strictly followed, the submission will not be accepted.**
- **If cheating is detected, all members of the teams involved will fail the course, both those who cheated and those who were cheated. They should also consult university regulations to consider other consequences.**
- **If AI tools are used in the project, the report must include a detailed explanation of the purpose for which the AI was used and the instructions given to obtain the results (screenshots of the process and interaction must be included). Failure to do so is a violation of academic honesty policies.**